

Alphaforge

Bitcoin Alpha Generation via On-Chain Macro ML Modelling

Finance and Analytics Club

Abstract—This project investigates the prediction of daily Bitcoin log returns using a combination of blockchain on-chain metrics, macroeconomic indicators, and historical price-derived features. Machine learning models, including Random Forest and XGBoost, are trained on historical data spanning 2014–2026 to forecast future returns. These forecasts are subsequently converted into long-short trading signals to simulate an alpha-generating strategy. The proposed approach is evaluated using both statistical forecasting metrics and financial performance measures such as Sharpe Ratio, Maximum Drawdown, and cumulative returns.

I. INTRODUCTION

Bitcoin is a highly volatile financial asset whose price is influenced by both blockchain-specific fundamentals and broader macroeconomic conditions. Unlike traditional financial assets, Bitcoin provides transparent on-chain data that can capture network usage, miner activity, and user adoption in real time. Simultaneously, macroeconomic variables such as the US Dollar Index and Treasury yields significantly affect investor risk appetite.

The objective of this project is to determine whether combining these heterogeneous information sources enables machine learning models to generate predictive signals capable of outperforming a passive buy-and-hold Bitcoin investment.

II. DATASET DESCRIPTION

The dataset consists of approximately 4,300 daily observations spanning September 2014 through 2026.

The original dataset contains

- Daily OHLCV market data
- Blockchain on-chain metrics
- Macroeconomic indicators

The target variable is the continuously compounded daily Bitcoin return

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right)$$

where P_t denotes the daily closing price.

While the original Rhode feature set forms the foundation of the predictive model, it was augmented with additional technical indicators to capture short-term temporal dependencies in Bitcoin price dynamics. The final feature set consists of 17 explanatory variables, combining blockchain-derived metrics, macroeconomic indicators, lagged returns, momentum measures, and rolling volatility estimates.

III. EXPLORATORY DATA ANALYSIS

Exploratory data analysis was conducted to understand the statistical properties of the dataset before model development.

The following analyses were performed

- Distribution plots for every predictor and the target variable
- Correlation heatmap between all Rhode features
- Stationarity testing using the Augmented Dickey-Fuller (ADF) test

Distribution plots revealed that several blockchain variables exhibit significant right-skewness, while macroeconomic variables demonstrate relatively smoother distributions. Correlation analysis indicated moderate relationships among valuation-based metrics while avoiding severe multicollinearity.

The ADF test showed that engineered return-based variables are largely stationary, making them suitable for predictive modelling.

TABLE I
AUGMENTED DICKEY-FULLER (ADF) STATIONARITY TEST RESULTS

Variable Description	<i>p</i> -value	Status
True Market Capitalization	0.6458	Non-Stationary
Total Fees (USD)	0.0000	Stationary
Fee-to-Market Cap Ratio	0.0000	Stationary
US Dollar Index (DXY) Close	0.0272	Stationary
10-Year Treasury Yield (TNX)	0.8371	Non-Stationary
Active Address Count	0.0951	Non-Stationary
Network Hash Rate	0.9991	Non-Stationary
Rolling Volatility	0.0001	Stationary
Return (Lag 1)	0.0000	Stationary
Return (Lag 2)	0.0000	Stationary
Return (Lag 3)	0.0000	Stationary
Return (Lag 5)	0.0000	Stationary
Return (Lag 7)	0.0000	Stationary
Return (Lag 14)	0.0000	Stationary
7-day Momentum	0.0000	Stationary
30-day Momentum	0.0000	Stationary
7-day Volatility	0.0000	Stationary
Log Returns	0.0000	Stationary

(Figure 2: Feature Distributions) *see last*

IV. FEATURE ENGINEERING

Although the Rhode feature set captures the underlying economic state of the Bitcoin network and macroeconomic environment, cryptocurrency returns exhibit strong temporal dependence that cannot be fully represented by static daily variables. To capture these dynamics, several additional technical features were engineered.

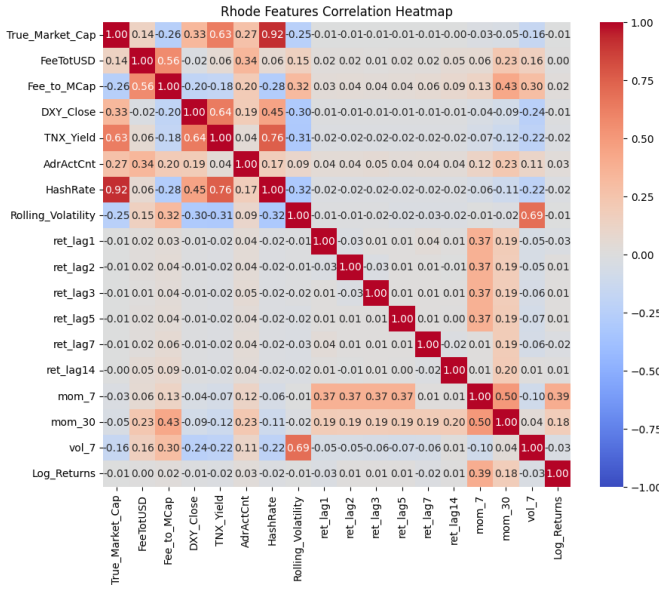


Fig. 1. Correlation matrix of the Rhode features.

A. Lagged Returns

Historical log returns from the previous 1-day, 2-day, 3-day, 5-day, 7-day, 14-day were included to capture autocorrelation and short-term momentum effects commonly observed in financial time series.

B. Momentum Indicators

Two momentum features were computed:

- 7-day momentum
- 30-day momentum

These quantify recent price appreciation and help distinguish short-term trends from longer market cycles.

C. Volatility Indicators

Market uncertainty was represented through

- 7-day rolling volatility
- 30-day rolling volatility

Rolling volatility captures changing market risk and allows the learning algorithms to adapt to varying volatility regimes.

V. DATA PREPROCESSING

Missing observations were forward-filled before removing incomplete rows.

The dataset was partitioned chronologically into

- Training (70%)
- Validation (15%)
- Testing (15%)

Feature scaling was performed using StandardScaler fitted exclusively on the training data.

VI. MACHINE LEARNING MODELS

Two ensemble learning algorithms were implemented.

A. Random Forest

Random Forest served as a nonlinear baseline model.

B. XGBoost

The XGBoost model was selected as the primary predictive model due to its strong performance on structured tabular datasets.

To improve feature utilisation and prevent excessive sparsity in feature importance estimates, several hyperparameters were adjusted:

- Number of estimators: 300
- Learning rate: 0.05
- Maximum tree depth: 4
- Row subsampling: 0.8
- Feature subsampling: 0.8
- Split penalty (γ) removed
- L1 regularisation removed

These modifications encouraged the model to utilise a broader range of explanatory variables while maintaining generalisation performance.

VII. ALPHA SIGNAL GENERATION

Predicted daily log returns were converted into binary trading signals. Positive predictions generated long positions, while negative predictions generated short positions. To eliminate look-ahead bias, trading signals were shifted by one trading day, ensuring that positions were opened only after the corresponding prediction became available.

- Positive prediction $\rightarrow$ Long
- Negative prediction $\rightarrow$ Short

Signals were shifted by one day to eliminate look-ahead bias before calculating strategy returns.

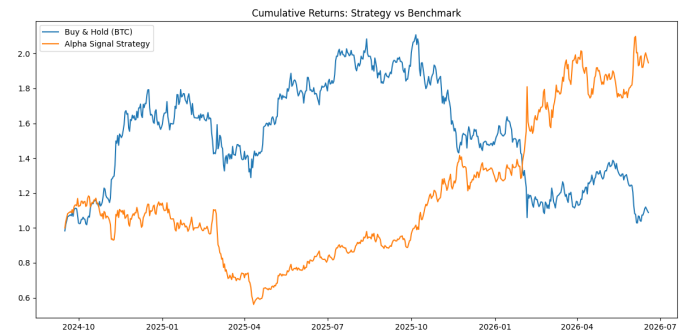


Fig. 3. Strategy vs Buy-and-Hold Performance

VIII. EVALUATION

Following are the results of the evaluations performed

TABLE II
FINANCIAL PERFORMANCE EVALUATION RESULTS

Metric	Value	Benchmark
Strategy Sharpe	0.84	0.11
Strategy Max Drawdown	-52.58%	-51.21%
Strategy Cum. Return	94.72%	8.81%

TABLE III
STATISTICAL PERFORMANCE EVALUATION RESULTS

Metric	Value
MAE	0.014259
RMSE	0.019336
R ²	0.337284
Directional Accuracy	66.41%

IX. FEATURE IMPORTANCE

Feature importance was analysed using three complementary techniques in order to reduce the bias associated with any single importance measure.

- **XGBoost Gain Importance** measures the average reduction in prediction loss contributed by each feature during tree construction
- **Split Frequency** measures how frequently each feature is selected for splitting across the ensemble.
- **Permutation Importance** provides a model-agnostic estimate of feature relevance by measuring the decrease in predictive accuracy after randomly shuffling each feature on the test set.

Using all three approaches provides a more reliable interpretation of the relative contribution of blockchain, macroeconomic, and engineered technical variables than relying on a single importance metric.

X. CONCLUSION

This study demonstrates that combining blockchain on-chain metrics with macroeconomic indicators and engineered technical features enables machine learning models to generate informative predictive signals for Bitcoin returns.

Although predicting cryptocurrency returns remains challenging, the proposed framework successfully integrates multiple sources of financial information into a systematic quantitative trading strategy.

XGBoost Feature Importance — On-Chain vs Macro Signals
(3 complementary methods)

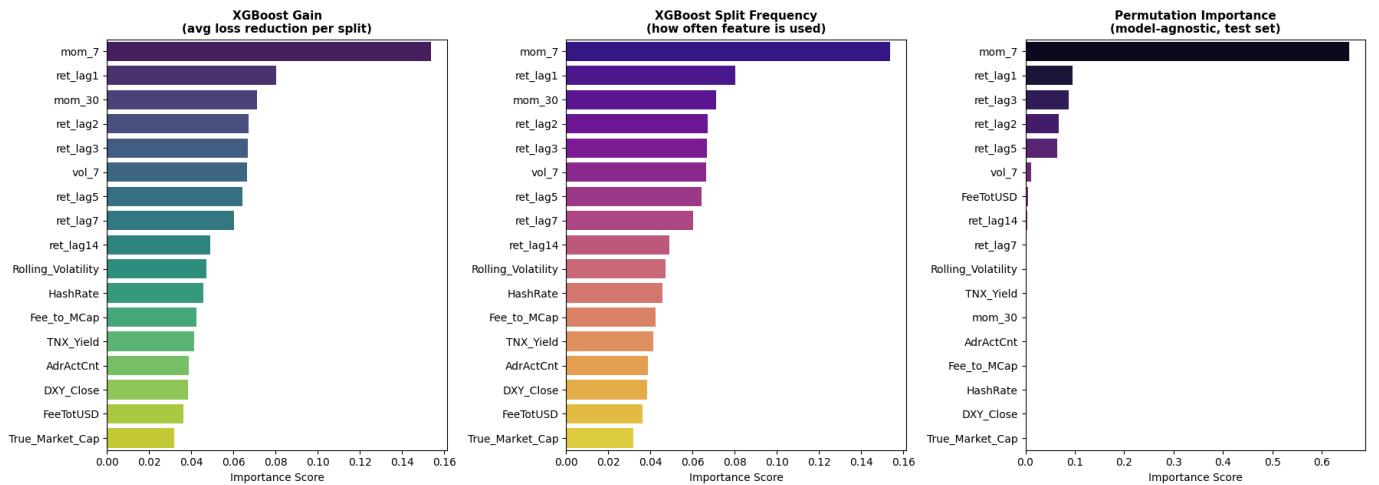


Fig. 4. Feature Importance Metrics Across Multi-Source Financial Indicators.

Feature Distributions

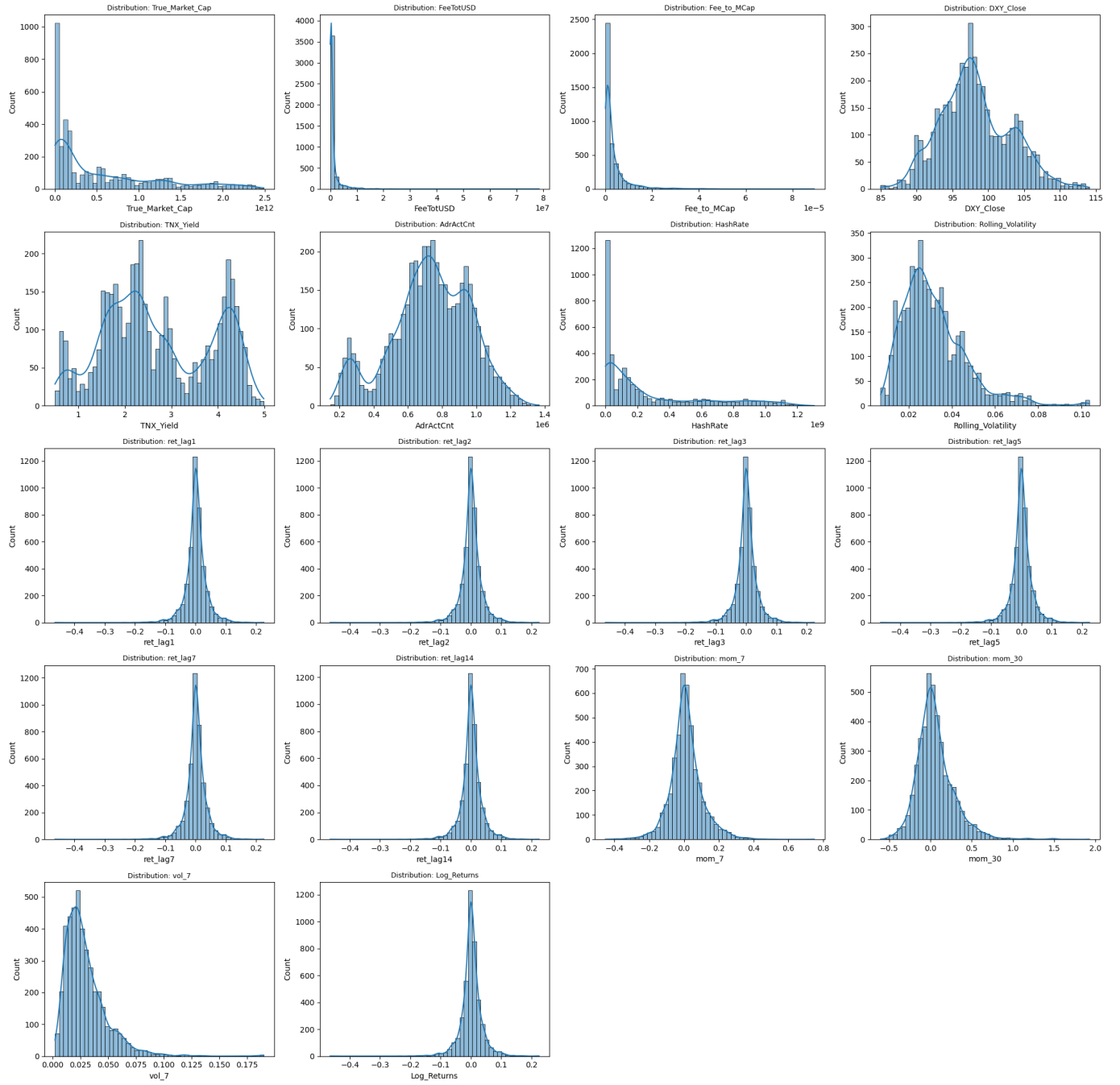


Fig. 2. Feature Distributions